

HUSSEINI SULEIMAN

2025/FWSD/11478

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COURSE CODE: SBT-DF202 • PRACTICAL LABORATORIES

ASSIGNMENT: Lab 1 — Digital Forensics Case Handling, Autopsy and Sleuth Kit Analysis

our practical context

You are acting as a junior digital forensic analyst in the ICdfa laboratory. You have been provided with a forensic disk image named Ch01InChap01.dd. The requesting officer requires you to identify deleted files, recover relevant office documents, perform keyword searches, preserve your analysis steps, verify evidence integrity, and prepare a repeatable technical forensic report. The original evidence image must not be modified during the investigation.

OBJECTIVES

What you must demonstrate

By completing this lab, trainees will demonstrate their ability to apply the digital forensic investigation process in a controlled practical environment. Students will preserve and hash a forensic disk image, perform structured analysis using Autopsy, examine the image using The Sleuth Kit command-line tools, identify and recover deleted files, analyse file metadata and data units, perform keyword searches, verify recovered evidence using cryptographic hashes, and prepare a professional forensic report containing repeatable findings.

Students are expected to demonstrate practical competence with tools and commands including Autopsy, img_stat, mmls, fsstat, fls, istat, icat, blkcat, blkls, tsk_recover, md5sum and sha256sum.

PROCEDURE

Laboratory instructions

Before beginning this lab, review the **Module 1 and Module 2 lessons and study materials** covering digital investigation procedures, evidence handling, chain of custody, forensic imaging, Autopsy and The Sleuth Kit.

Download the authorised training forensic image [Ch01InChap01.dd](#) and create a dedicated forensic working directory before beginning your analysis. If the direct download does not start automatically, use the [browser download page](#) instead. The source lab specifically provides this evidence image for the practical.

After downloading the evidence image, record the evidence source, file size, storage location, MD5 hash and SHA-256 hash before conducting any forensic analysis. Preserve the original image and perform recovery and examination activities in separate working folders.

ATTEMPT TO THE LABS

FILE INFORMATIONS:

File name: **Ch01InChap01.dd**

File size: 1440KB

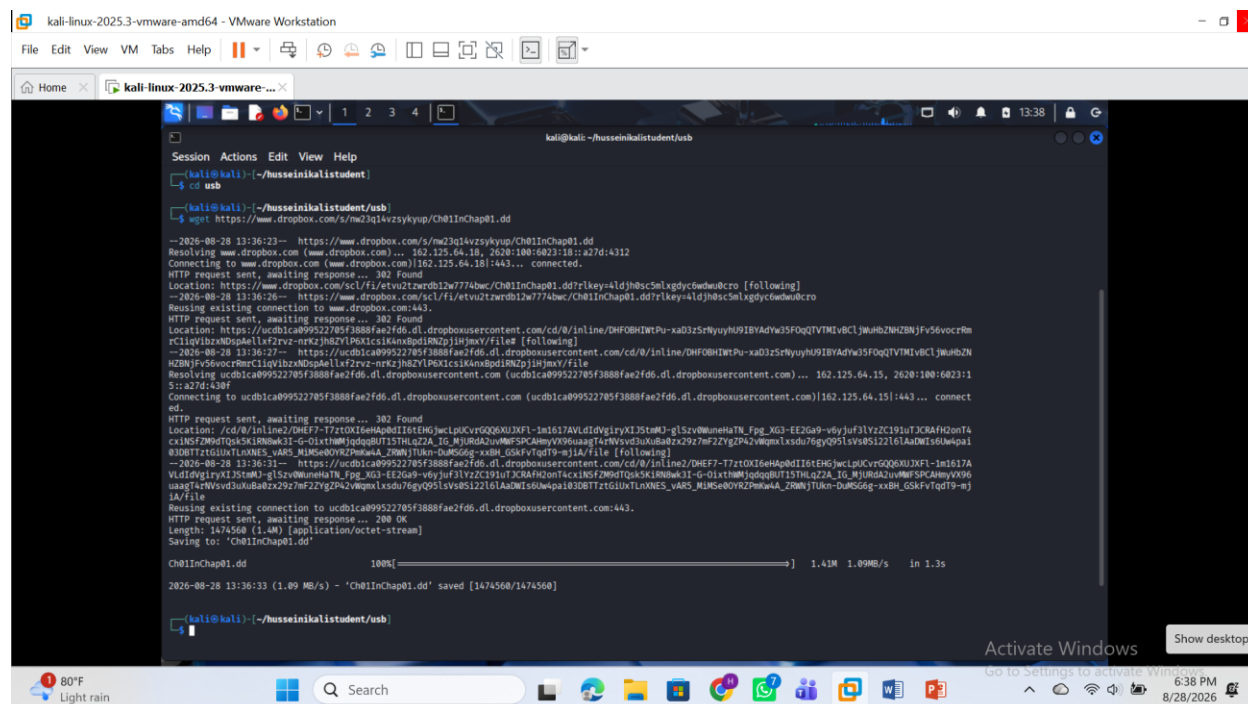
Storage part: C:\INVESTIGATION FILE\ **Ch01InChap01.dd** ON WINDOWS

(kali@kali)-[~/husseinikalistudent/usb ON KALI

MD5HASH VALU: a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd

Part A — Autopsy Analysis

Create a forensic case in Autopsy and add **Ch01InChap01.dd** as the evidence source. Identify deleted files, recover relevant files, perform keyword searches, tag relevant evidence and generate an Autopsy report. Record the case name, evidence source, deleted-file findings, recovered files, keyword-search results and reporting evidence.



```
kali@kali: ~/husseinikalistudent/usb
wget https://www.dropbox.com/s/m23q14vzsykyup/Ch01InChap01.dd
--2026-08-28 13:36:23-- https://www.dropbox.com/s/m23q14vzsykyup/Ch01InChap01.dd
Resolving www.dropbox.com (www.dropbox.com)... 162.125.64.15, 2620:100:6021:18::a27d:4312
Connecting to www.dropbox.com (www.dropbox.com)[162.125.64.15]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/SCl/Fi/etvu2tzwrdb12w774bwc/Ch01InChap01.dd?rlkey=a1djh8cSmxgdyrcbuehu@cro
--2026-08-28 13:36:26-- https://www.dropbox.com/SCl/Fi/etvu2tzwrdb12w774bwc/Ch01InChap01.dd?rlkey=a1djh8cSmxgdyrcbuehu@cro
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://ucd81ca899522705f3888fae2fd6.dl.dropboxusercontent.com/cd/0/inline/DHFOBHlWtPu-xaD3z5rWuyyH9I8YAdYw35FoQqTVtMIvBCLjWuMbZnH2BNjFv56vocrRm
K1lqVibznDopAellk7rVz2-nrkZjH8ZVlPKXlcaKxnpdRNzpjHjmxv/Files [following]
--2026-08-28 13:36:27-- https://ucd81ca899522705f3888fae2fd6.dl.dropboxusercontent.com/cd/0/inline/DHFOBHlWtPu-xaD3z5rWuyyH9I8YAdYw35FoQqTVtMIvBCLjWuMbZn
H2BNjF56vocrRmK1lqVibznDopAellk7rVz2-nrkZjH8ZVlPKXlcaKxnpdRNzpjHjmxv/Files
Resolving ucd81ca899522705f3888fae2fd6.dl.dropboxusercontent.com (ucd81ca899522705f3888fae2fd6.dl.dropboxusercontent.com)... 162.125.64.15, 2620:100:6021:1
5:a27d:430f
Connecting to ucd81ca899522705f3888fae2fd6.dl.dropboxusercontent.com (ucd81ca899522705f3888fae2fd6.dl.dropboxusercontent.com)[162.125.64.15]:443... connect
ed.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DHEF7-T7zt0XIGeHap0II6tEH6wclPUKvGQ6XJXFL-jm1617A
Vld0g9irYXl3stMj-q15zVWumehaTN_Fpg_KG3-EE2Ga9-vdyjuf31Vz2C19tJ2X8AFHqonT4
c1MSFZM9dQKsKRNhwk3I-q1xttMmYqogB1T3THLc2ZA_Tc_HJUR82uWmSPCdmYx96
uaag74Wv9yd3uM8aBz29z7Mf2YgZD42VgmXksdu76gyQ951SVS85122161AaDwIs8uWpa18308TTz1G1UxTLnXNES_vAR5_MUMSe0YvR2PmAwLA_ZMNjTUKn-DUMSG6g-xxBh_6SkFvTqdT9-mj1A/Files [following]
--2026-08-28 13:36:31-- https://ucd81ca899522705f3888fae2fd6.dl.dropboxusercontent.com/cd/0/inline2/DHEF7-T7zt0XIGeHap0II6tEH6wclPUKvGQ6XJXFL-jm1617A
Vld0g9irYXl3stMj-q15zVWumehaTN_Fpg_KG3-EE2Ga9-vdyjuf31Vz2C19tJ2X8AFHqonT4c1MSFZM9dQKsKRNhwk3I-q1xttMmYqogB1T3THLc2ZA_Tc_HJUR82uWmSPCdmYx96
uaag74Wv9yd3uM8aBz29z7Mf2YgZD42VgmXksdu76gyQ951SVS85122161AaDwIs8uWpa18308TTz1G1UxTLnXNES_vAR5_MUMSe0YvR2PmAwLA_ZMNjTUKn-DUMSG6g-xxBh_6SkFvTqdT9-mj1A/Files
Reusing existing connection to ucd81ca899522705f3888fae2fd6.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1474560 (1.4M) [application/octet-stream]
Saving to: 'Ch01InChap01.dd'

Ch01InChap01.dd      100% [=====] 1.41M  1.09MB/s   in 1.3s

2026-08-28 13:36:33 (1.09 MB/s) - 'Ch01InChap01.dd' saved [1474560/1474560]

(kali@kali)-[~/husseinikalistudent/usb]
```

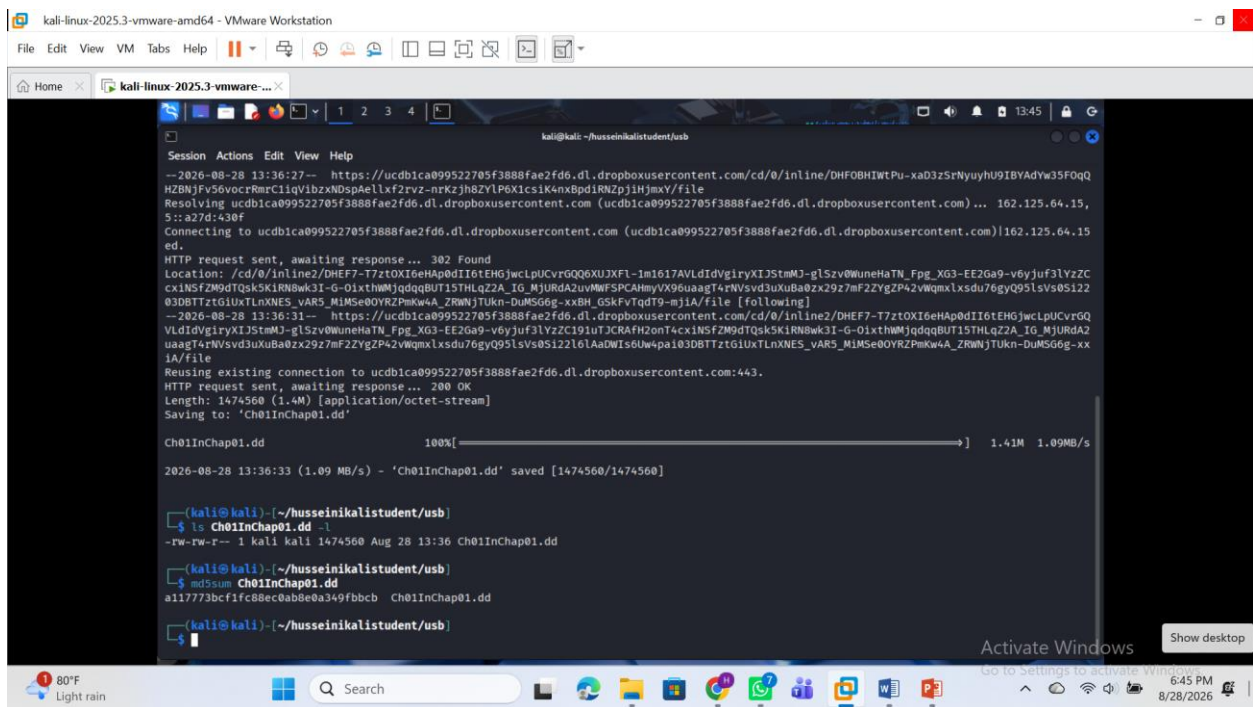
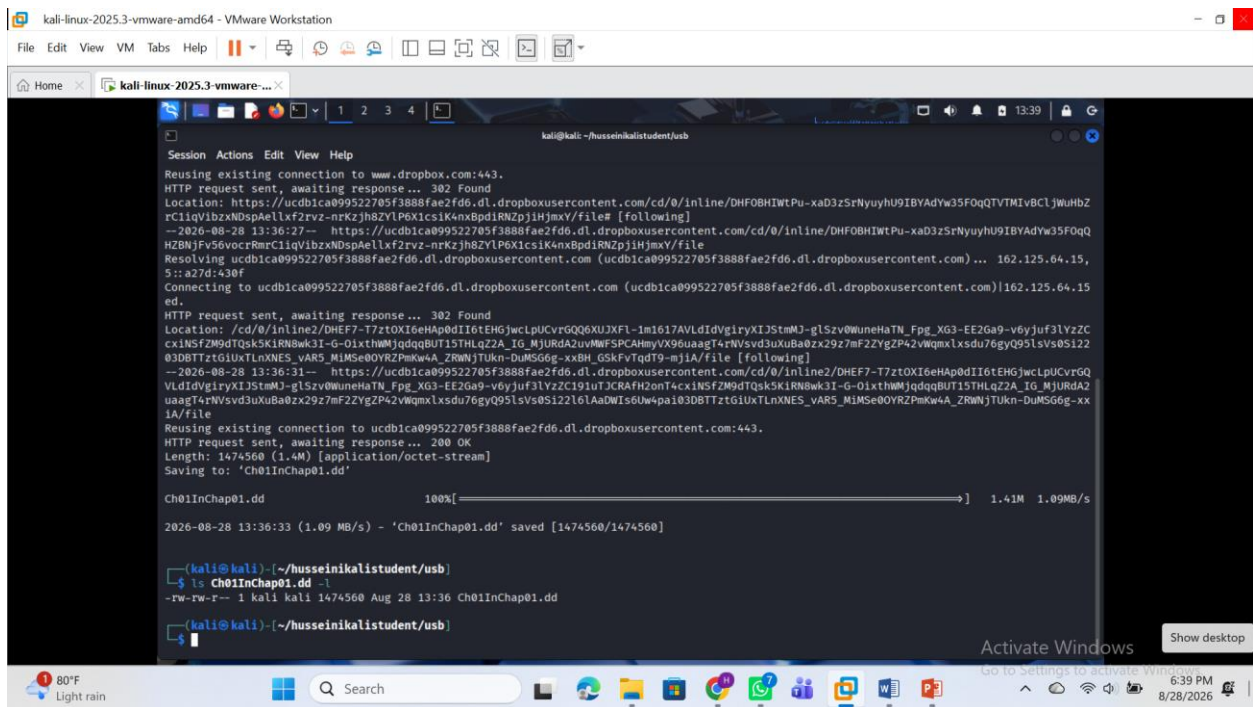


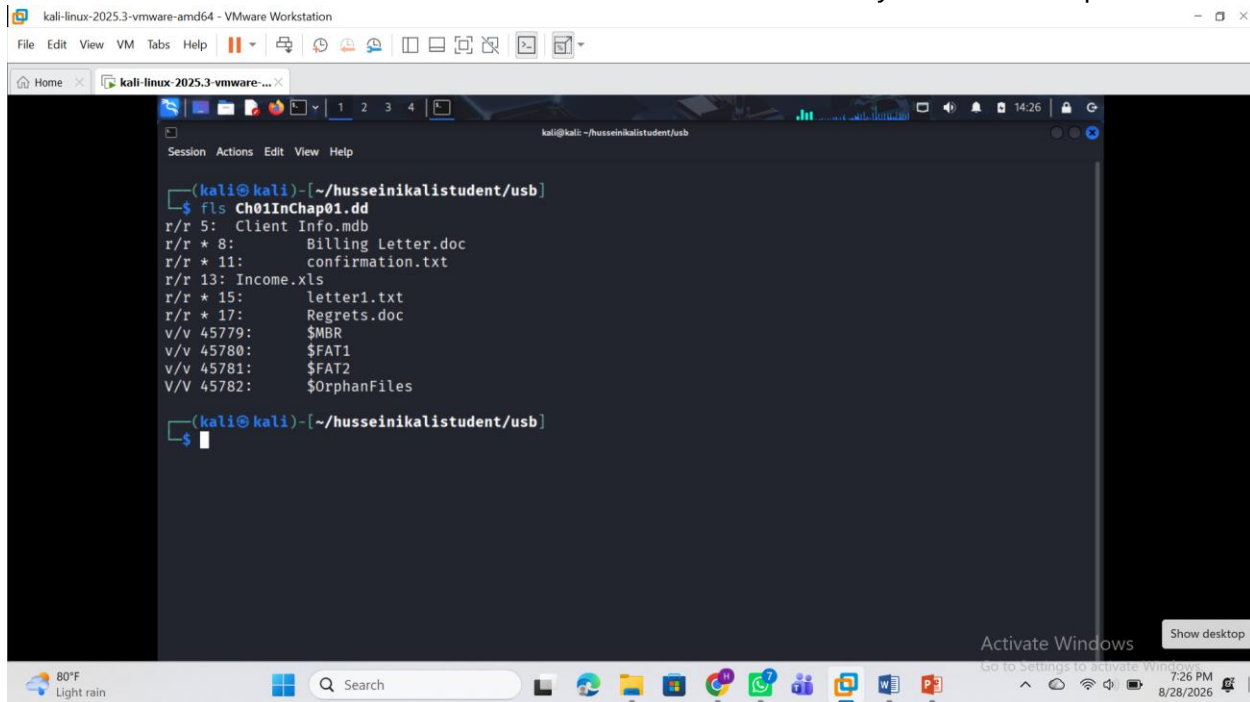
IMAGE TYPE AND SIZE: Image Type: raw

Size in bytes: 1474560

Sector size: 512

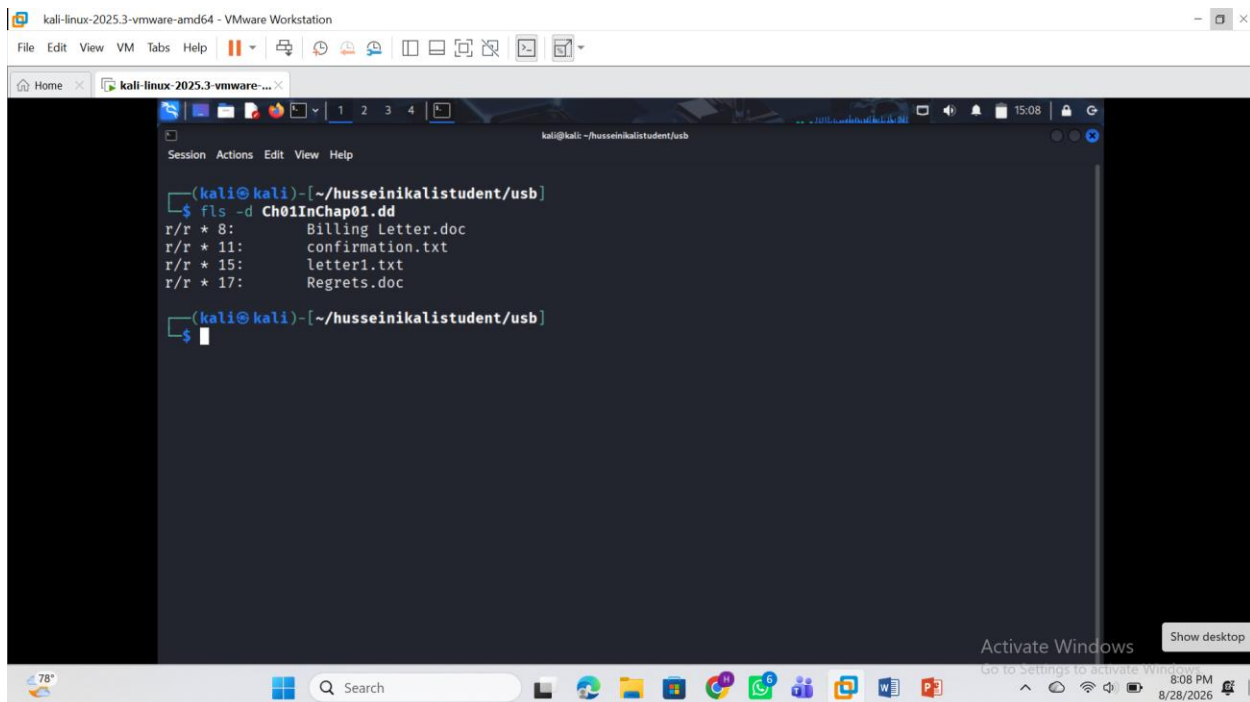
IDENTIFICATION OF DELETED FILES AND DIRECTORY AND ENTRIES

Command use to list List allocated and deleted file names in a directory: `fls Ch01InChap01.dd`



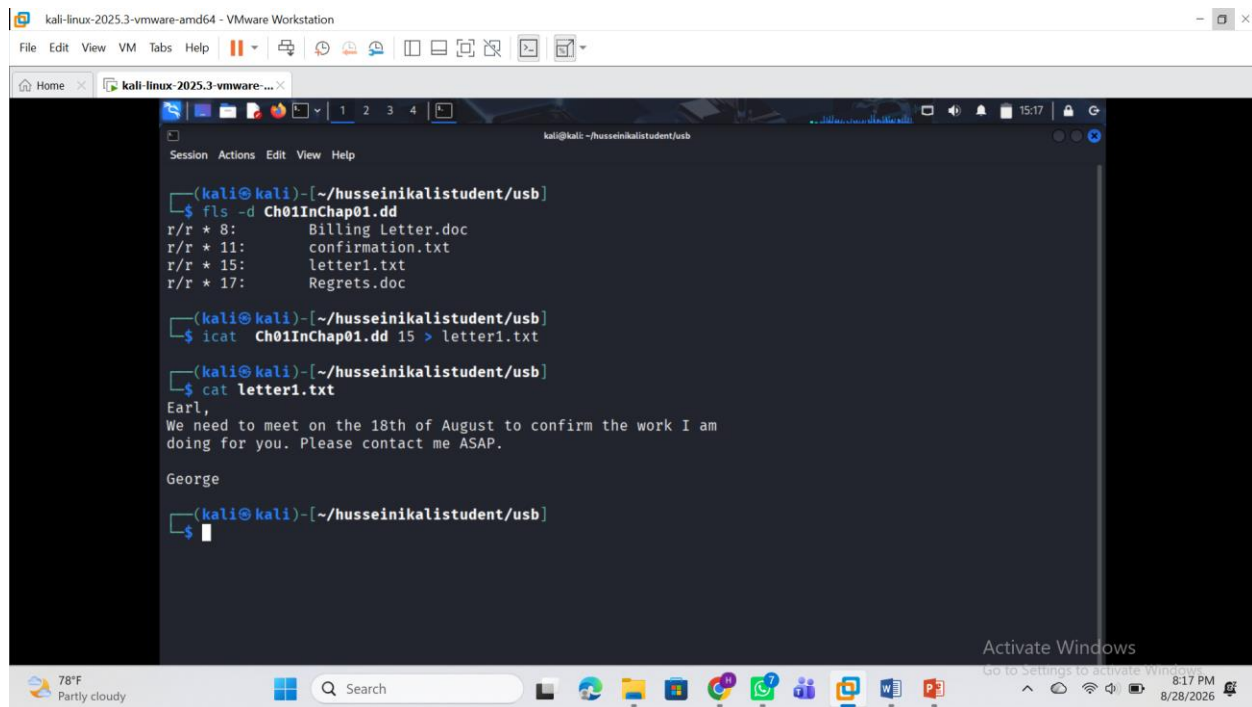
```
kali@kali: ~/husseinikalistudent/usb
$ fls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

`fls -d Ch01InChap01.dd` : Is the Command used to Display deleted entries only



```
kali@kali: ~/husseinikalistudent/usb
$ fls -d Ch01InChap01.dd
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
```

Recover deleted files using *inode* without specifying a file name



```
(kali@kali)-[~/husseinikalistudent/usb]
$ ls -l Ch01InChap01.dd
-r/r * 8:      Billing Letter.doc
-r/r * 11:     confirmation.txt
-r/r * 15:     letter1.txt
-r/r * 17:     Regrets.doc

(kali@kali)-[~/husseinikalistudent/usb]
$ icat Ch01InChap01.dd 15 > letter1.txt

(kali@kali)-[~/husseinikalistudent/usb]
$ cat letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(kali@kali)-[~/husseinikalistudent/usb]
$
```

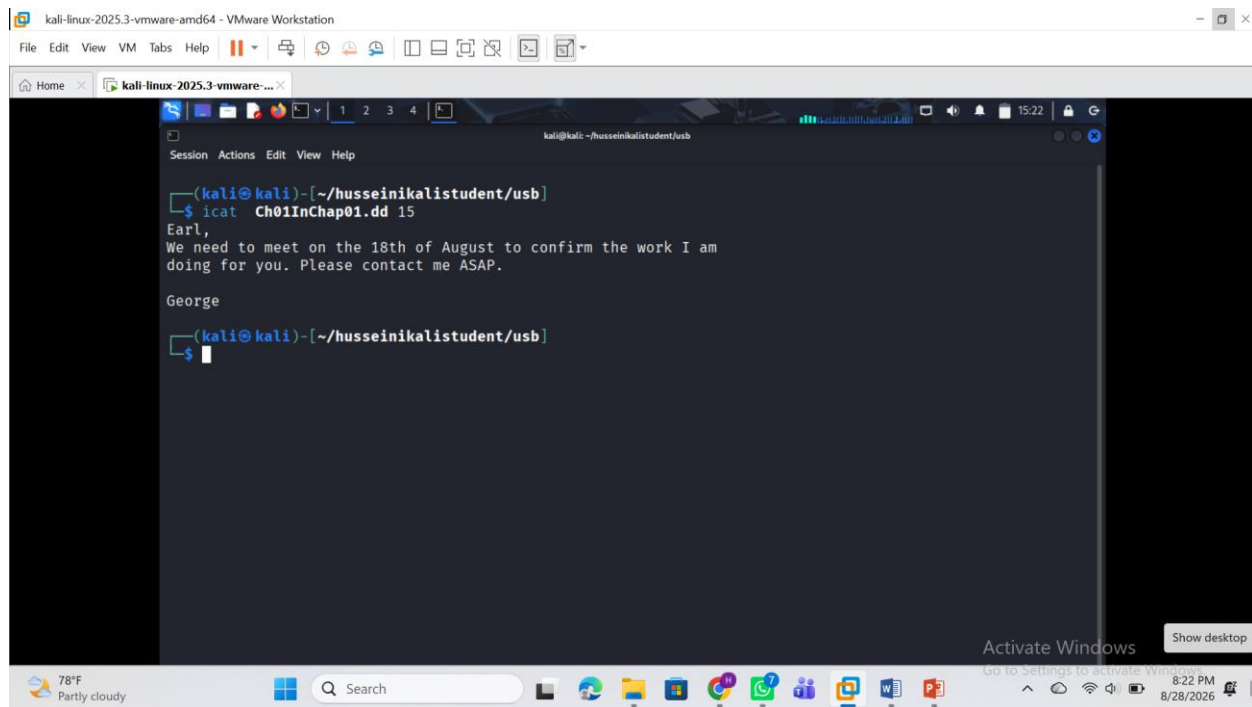
Comand used:

```
icat Ch01InChap01.dd 15 > letter1.txt
```

```
cat letter1.txt
```

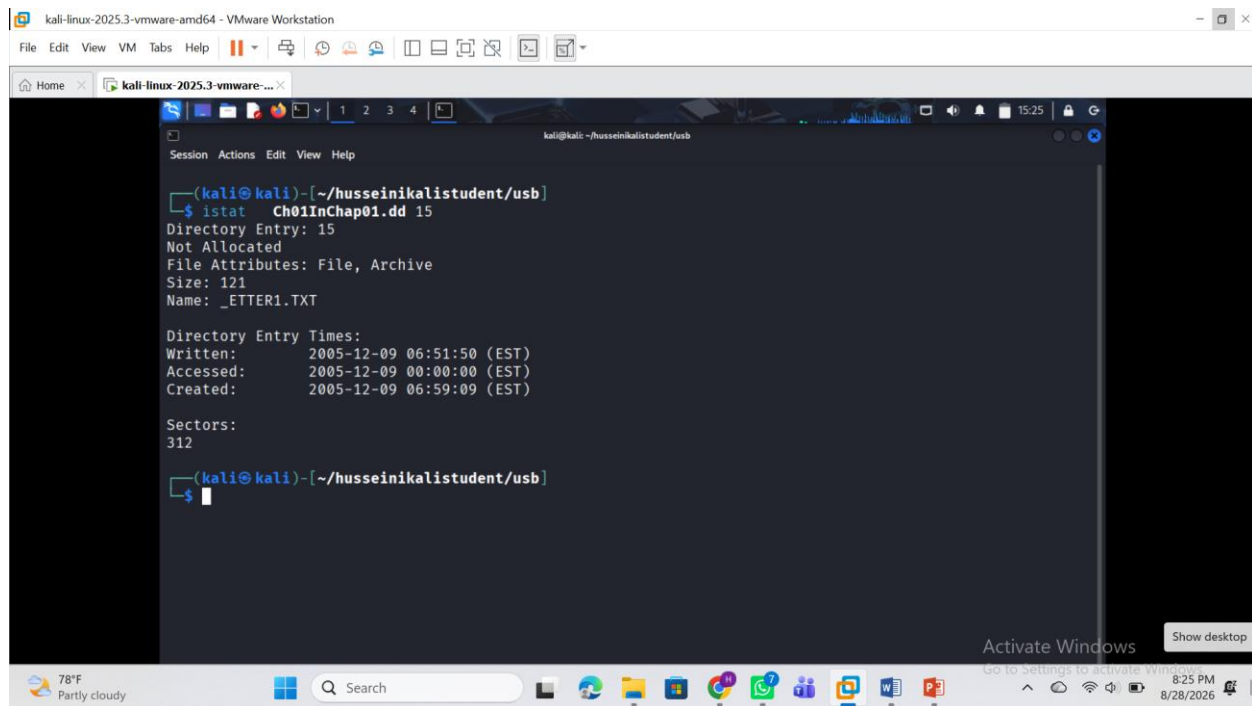
Extract the data units of letter1.txt (inode15)

See attach



Display the statistics and details about a given meta data structure

Command used: istat Ch01InChap01.dd 15



Part B — Sleuth Kit Analysis

Examine the forensic image using `img_stat`, `mmls`, `fsstat` and `fls`. Identify deleted files and locate **INCOME.XLS**. Determine its metadata address, examine its metadata and data units using `istat`, recover it using `icat` and `blkcat` where applicable, perform additional recovery using `tsk_recover`, and compare the hashes of the recovered copies.

Document every significant command, output, observation, error and corrective action.

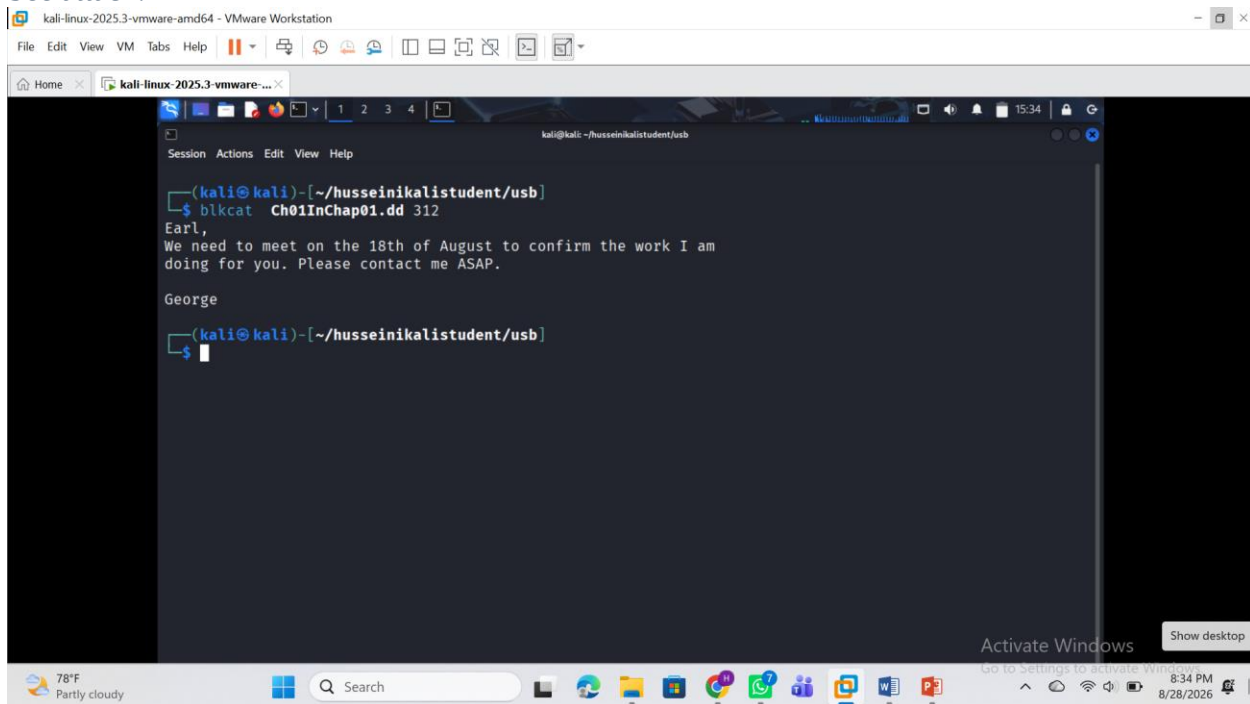
Screenshots must clearly support the work claimed in your report.

Attempt:

Extract the contents of a given data unnnits

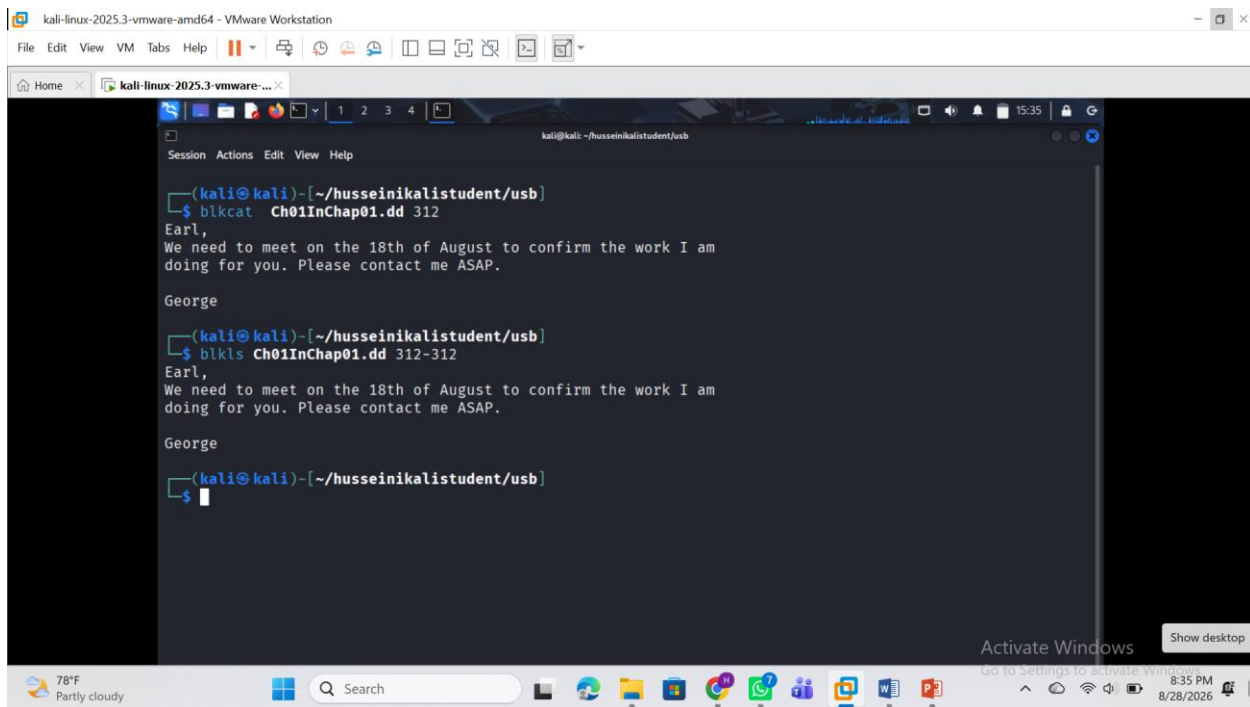
Command used: `blkcat Ch01InChap01.dd 312`

See attach:

A screenshot of a Kali Linux terminal window running inside a VMware Workstation. The terminal shows the command `blkcat Ch01InChap01.dd 312` being executed. The output of the command is a text message: "Earl, We need to meet on the 18th of August to confirm the work I am doing for you. Please contact me ASAP." followed by a new line and the name "George". The terminal prompt is `(kali@kali)~[/husseinikalistudent/usb]`. The VMware window title is "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The Windows taskbar at the bottom shows the date as 8/28/2026 and the time as 8:34 PM.

Command used: `blkls Ch01InChap01.dd 312-312:`

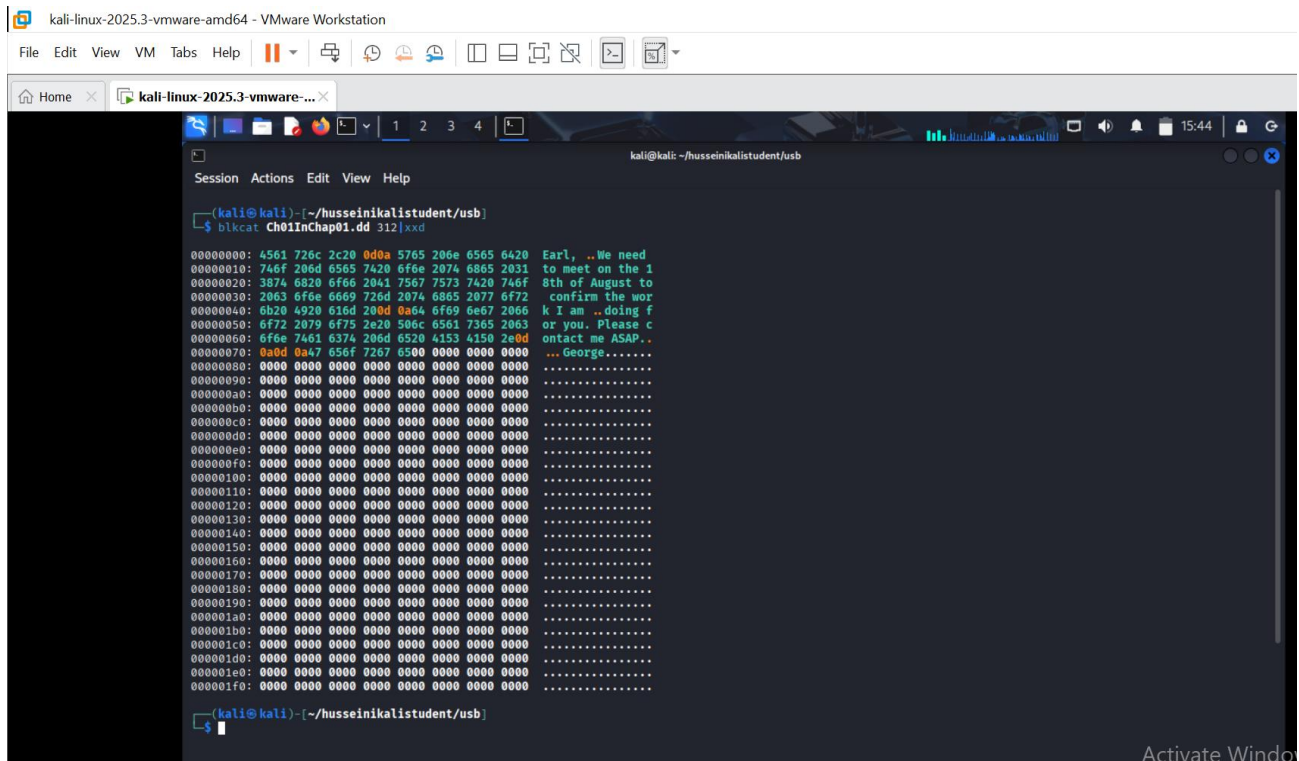
For: List the details about data units and can extract the unallocated space of the file system



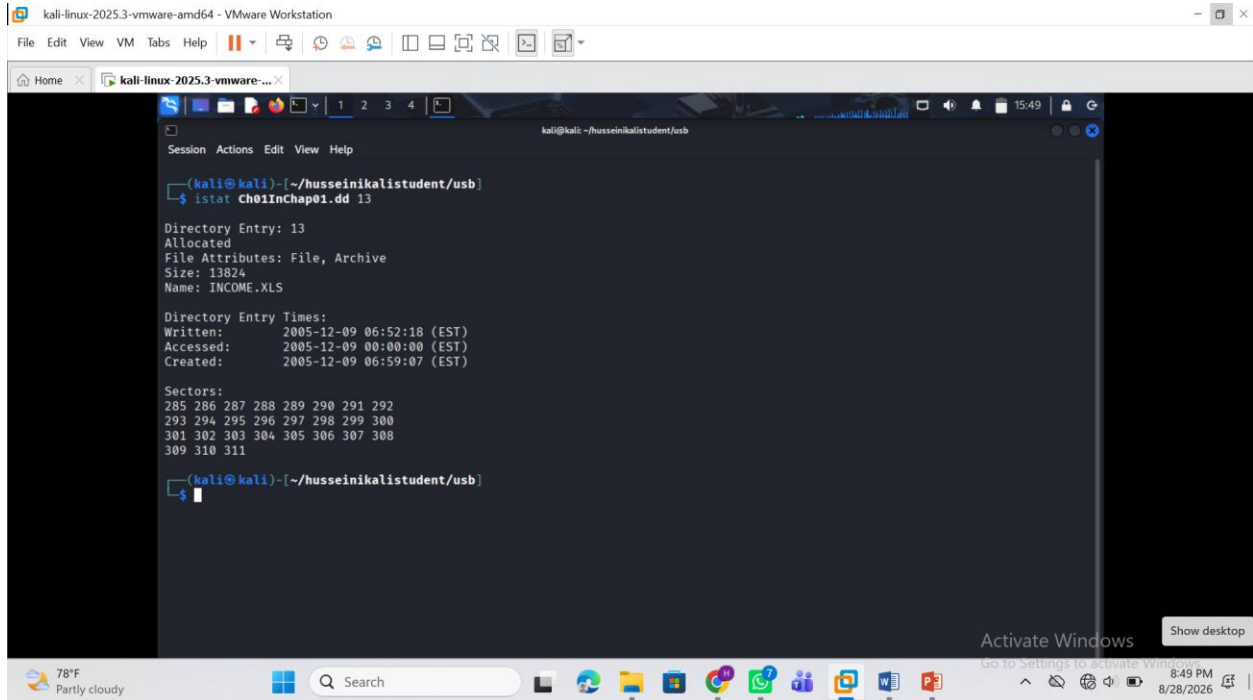
Extracting the contents of a given data unit in Hex

Command used: blkcat Ch01InChap01.dd 312|xxd

See attach:



Show A file with multiple sectors



The screenshot shows a terminal window titled 'kali@kali: ~/husseinikalistudent/usb'. The user has entered the command `lsstat Ch01InChap01.dd 13`. The output displays file attributes and sector information.

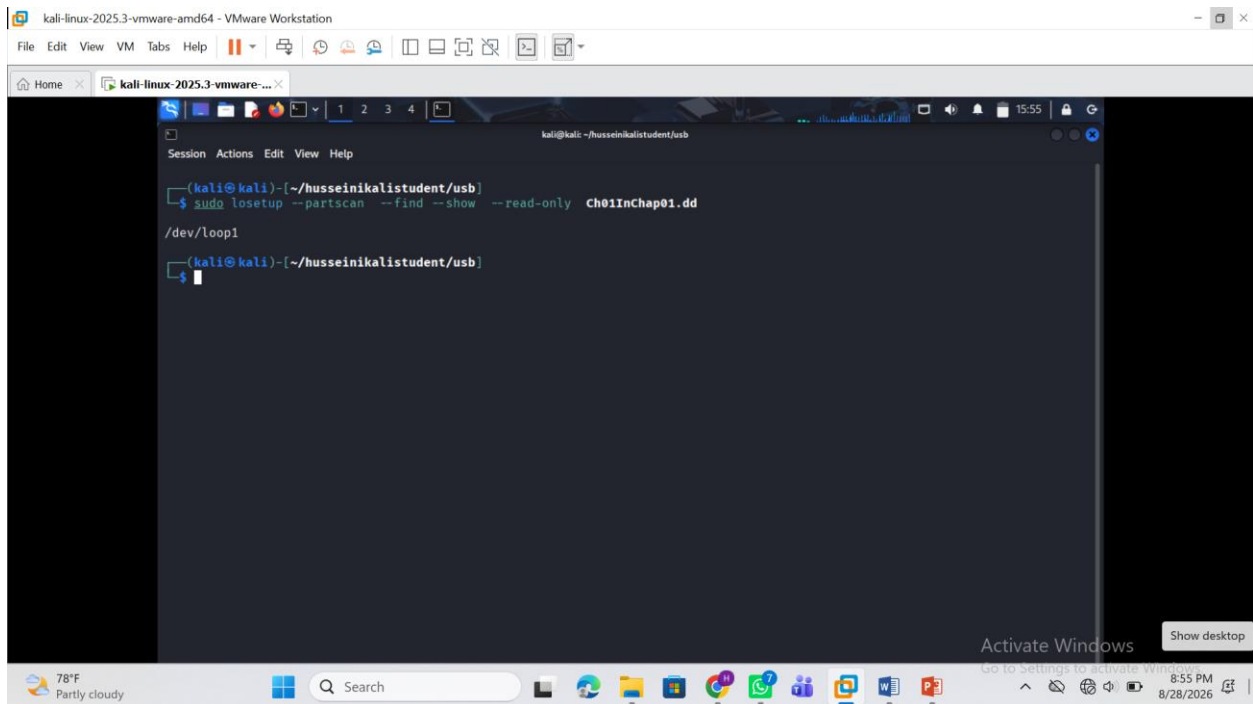
```
(kali@kali)~/husseinikalistudent/usb
$ lsstat Ch01InChap01.dd 13

Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written: 2005-12-09 06:52:18 (EST)
Accessed: 2005-12-09 00:00:00 (EST)
Created: 2005-12-09 06:59:07 (EST)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311
```

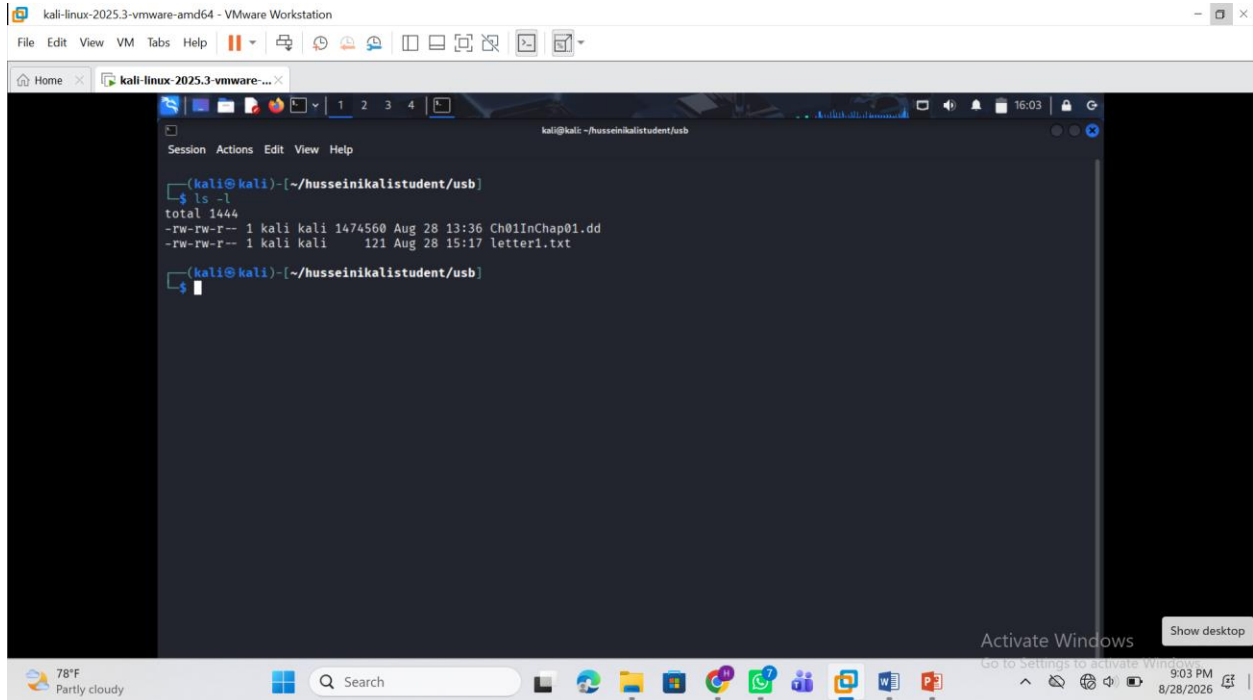
Mounting a USB image



The screenshot shows a terminal window titled 'kali@kali: ~/husseinikalistudent/usb'. The user has entered the command `sudo losetup --partscan --find --show --read-only Ch01InChap01.dd`. The output shows the device `/dev/loop1` has been created.

```
(kali@kali)~/husseinikalistudent/usb
$ sudo losetup --partscan --find --show --read-only Ch01InChap01.dd

/dev/loop1
(kali@kali)~/husseinikalistudent/usb
```



Do not modify the original forensic image. All recovered evidence, screenshots, command outputs and reports must be stored separately from the source image.

Required evidence

Students must submit evidence demonstrating completion of the practical, including:

1. Dedicated lab folder structure.
2. Evidence image filename, size and storage location.
3. MD5 and SHA-256 hashes of Ch01InChap01.dd.
4. Completed mini chain-of-custody/evidence worksheet.
5. Autopsy case creation and evidence source configuration.
6. Autopsy deleted-file listing.
7. Autopsy keyword-search results.
8. Tagged/recovered Autopsy evidence and generated report.
9. img_stat, mmls/fsstat and fls outputs.
10. Recursive deleted-file listing using fls.
11. Identification of the metadata address for INCOME.XLS.
12. INCOME.XLS recovery using icat.
13. istat output showing file size and relevant data units.
14. blkcat recovery attempt and resulting evidence.
15. Hash comparison of recovered INCOME.XLS copies, including tsk_recover where applicable.
16. Evidence that the recovered spreadsheet was successfully examined or converted locally.

The final submission must include one professional PDF forensic report containing explanations

and screenshots, together with the required ZIP archive containing recovered files, reports and screenshot evidence. The original DD image must not be submitted unless specifically requested by the instructor.

Lab access notes This practical forms part of the first laboratory assessment window. Students must review the relevant Module 1 and Module 2 lessons and study materials before beginning. Use only the authorised training evidence supplied for this lab. Maintain evidence integrity throughout the exercise and do not modify the original DD image. Submission must include the required PDF forensic report and supporting evidence files.